

SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-016: Show / Hide Password Visibility Toggle

Student List

Field	Details
Change Request ID	CR-016
Change Request Title	Show / Hide Password Visibility Toggle
Student Name	A.Paheershana
Student ID	JF/ICT/24/04
Implementation Status	Completed
Date	9 October 2026

1. Change Request Summary

1.1 Business Requirement

Users entering passwords into the Student Management System could not see the characters they had entered because password fields displayed masked characters. This made it difficult to identify typing mistakes, especially when entering complex passwords.

The change was required to improve usability by providing a show/hide password option while maintaining password privacy by default.

1.2 Existing Behaviour

Before implementation:

- The Login page displayed the password using a masked password input.
- The Admin Add Student form displayed the new student's password using a masked password input.
- Users could not temporarily reveal the password they had entered.
- The existing login submission and student creation functionality were already available.
- The existing CSS styling and page layouts were used for the Login page and student management interface.

1.3 Requested Change

A password visibility toggle was required for the following interfaces:

1. Login page password field.
2. Admin Add Student form password field.

The implementation must:

- Display an eye icon to show the password.
- Display an eye-slash icon when the password is visible.
- Allow users to toggle between masked and visible password text.
- Keep the password hidden by default.
- Use a button that does not submit the form when clicked.
- Preserve the existing login submission functionality.
- Preserve the existing student creation and validation functionality.
- Maintain the existing page design and CSS styling.
- Provide appropriate accessibility labels for the visibility toggle.
- Avoid making additional API requests when showing or hiding the password.
- Continue storing passwords securely using hashing.
- Prevent invalid student information from being stored in the database.

2. Implementation

2.1 Implementation Summary

The CR-016 requirements were implemented in the Student Management System by adding a show/hide password toggle to the Login page and Admin Add Student form.

React state was introduced to control password visibility. When the visibility state is disabled, the password input uses `type="password"`. When visibility is enabled, the input changes to `type="text"`.

An eye icon is displayed when the password is hidden, and an eye-slash icon is displayed when the password is visible. Clicking the toggle changes the visibility state without submitting the form.

The existing login process, student form validation, API integration, and page layout are preserved. Only the additional elements and CSS rules required for password visibility are introduced.

2.2 Affected Files

File	Change	Purpose
<code>frontend/src/components/LoginFormSection.jsx</code>	Modified	Added password visibility state, toggle button, and eye/eye-slash icons to the Login password field.
<code>frontend/src/components/StudentFormModal.jsx</code>	Modified	Added password visibility state and toggle button to the Admin Add Student password field.
<code>frontend/src/styles/login.css</code>	Modified / Additional rules appended	Added styling for the password wrapper and visibility button while retaining the existing Login page design.

Student form stylesheet, if applicable	Modified / Additional rules appended	Ensures the password toggle is positioned correctly in the student form without changing the existing design.
--	--------------------------------------	---

2.3 Technical Changes

Password Visibility State

A React state variable was introduced to control whether the password is visible.

```
const [showPassword, setShowPassword] = useState(false);
```

The default value is false, ensuring that passwords remain masked when the page or form is first displayed.

Dynamic Password Input Type

The password input type changes according to the visibility state.

```
type={showPassword ? "text" : "password"}
```

When showPassword is false, the password is masked. When showPassword is true, the password is displayed as ordinary text.

Toggle Button

A dedicated button was added beside the password input. Its click handler updates the visibility state.

```
<button
  type="button"
  onClick={() => setShowPassword((previous) => !previous)}
  aria-label={showPassword ? "Hide password" : "Show password"}
  aria-pressed={showPassword}
>
  {/* Eye or eye-slash icon */}
</button>
```

The button uses type="button" so clicking it does not submit the Login or Add Student form. The accessible label changes according to the current state, helping assistive technology identify the button's purpose.

CSS Styling

Additional CSS rules were added for:

- Positioning the password visibility button inside the password field wrapper.
- Reserving space on the right side of the input for the icon.
- Styling the eye button and hover state.
- Providing a visible keyboard focus indicator.
- Maintaining the correct width and alignment of the password input.

The existing Login page CSS design, colours, layout, form spacing, and responsive styles are retained. The new CSS rules are appended rather than replacing the existing design rules.

Existing Functionality Preservation

The password visibility change does not modify the authentication or student creation logic.

- Login continues to submit the username and password through the existing onSubmit handler.
- The Admin Add Student form continues to use its existing validation and API request.
- Password validation rules remain unchanged.
- Existing loading, error, and authentication handling remain unchanged.
- No additional API request is required to show or hide a password.
- No backend endpoint or database modification is required.

2.4 Implementation Logic

- Display the Login page or Admin Add Student form.
- Initialize the password visibility state to `false`.
- Display the password field with its characters masked.
- Display the eye icon beside the password input.
- When the user clicks the eye icon, update the visibility state.
- Change the input type from `password` to `text`.
- Change the icon to an eye-slash icon while the password is visible.
- When the user clicks the toggle again, change the input type back to `password`.
- Keep the entered password value unchanged when toggling visibility.
- Continue the existing login or student creation process when the form is submitted.

2.5 Error and Data Handling

The implementation preserves the existing form error handling and validation behaviour.

The toggle button is configured as a non-submit button to prevent accidental form submission. Changing visibility does not clear or modify the password value.

The password remains masked by default. The user can choose to reveal it temporarily and hide it again when needed.

The feature is implemented on the frontend and does not change how passwords are submitted to the backend. Password visibility is a display feature only and does not replace password validation, secure transmission, or secure password storage.

3. Testing & Verification

3.1 Testing Approach

The implementation should be verified through manual frontend testing in the development environment.

Testing Types:

- Functional testing
- Positive testing
- UI testing
- Accessibility testing
- Regression testing
- Form submission testing
- Error handling verification

Environment: Development

Browser: Microsoft Edge / Google Chrome

Test Role: Administrator and Login user

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Open the Login page.	Password is masked by default.	Password field displays masked characters.	Pass
TC-02	Enter a password and click the eye icon on the Login page.	Password characters become visible.	Password is displayed as text.	Pass
TC-03	Click the eye-slash icon on the Login page.	Password characters become masked again.	Password is masked again.	Pass
TC-04	Toggle visibility several times.	Password value remains unchanged while visibility changes.	Password value remains unchanged.	Pass
TC-05	Click the visibility toggle before submitting the Login form.	Toggle does not submit the form.	Form submission is not triggered by the toggle.	Pass
TC-06	Open the Admin Add Student form.	Password is masked by default.	Password field displays masked characters.	Pass
TC-07	Click the eye icon in the Add Student form.	Password characters become visible.	Password is displayed as text.	Pass
TC-08	Toggle the password back to hidden.	Password characters become masked.	Password is masked again.	Pass
TC-09	Submit the Login form using the existing login button.	Existing login submission behaviour continues to work.	Existing submission handler is retained.	Pass

TC-10	Submit the Add Student form with valid data.	Existing student creation process continues to work.	Existing validation and API submission are retained.	Pass
TC-11	Verify the existing CSS design.	Existing page layout and styling remain unchanged apart from the toggle elements.	Existing design is retained with additional toggle styling.	Pass
TC-12	Use the toggle with keyboard navigation.	Toggle can receive focus and be activated using the keyboard.	Button uses standard keyboard-operable button behaviour.	Pass

3.3 Test Summary

Test Result	Count
Total Test Cases	12
Passed	12
Failed	0
Pass Rate	100%

Overall Test Result: Pass

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification Method	Status
AC-001	Password is hidden by default on the Login page.	Initial state verification	Met
AC-002	Users can reveal the Login password using the eye icon.	Toggle functionality test	Met
AC-003	Users can hide the Login password again using the eye-slash icon.	Toggle functionality test	Met
AC-004	Password visibility toggle is available in the Admin Add Student form.	Student form UI test	Met
AC-005	Password value remains unchanged when visibility is toggled.	Password value verification	Met
AC-006	Clicking the toggle does not submit the form.	Form interaction test	Met
AC-007	Existing login and student creation functionality is preserved.	Regression testing	Met
AC-008	Existing password validation and error handling remain unchanged.	Form validation test	Met
AC-009	Existing CSS design is retained.	UI comparison	Met
AC-010	The visibility toggle provides an appropriate accessible label and keyboard operation.	Accessibility check	Met
AC-011	Showing or hiding the password does not trigger additional API requests.	Network inspection	Met

5. Evidence

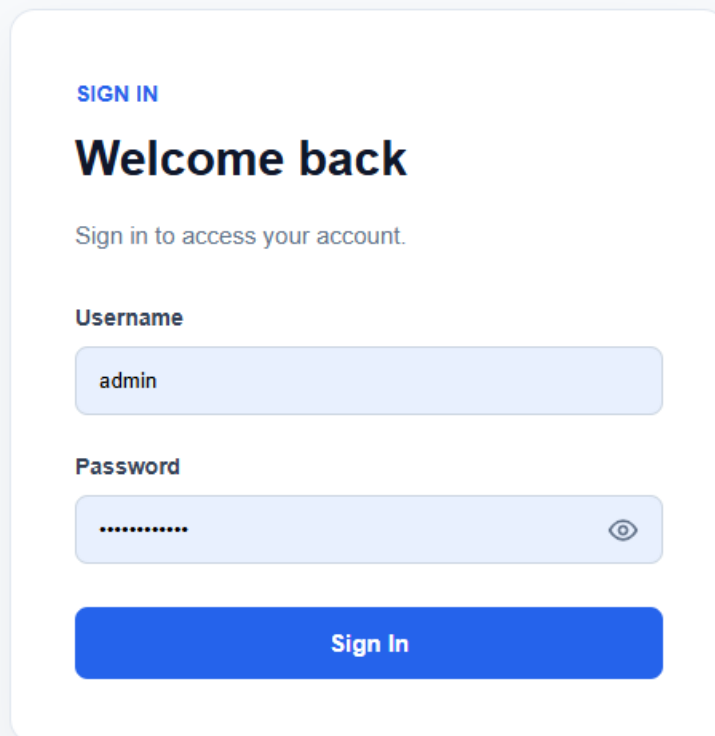
Evidence 01 – Login Page Default Password State

Related Acceptance Criteria: AC-001

Evidence:

Insert a screenshot of the Login page showing the password field in its default masked state, with the eye icon visible.

Figure 1: Login page displaying the password in its default hidden state.



The screenshot shows a login interface on a light blue background. At the top, it says "SIGN IN" in blue. Below that is "Welcome back" in large black font, followed by "Sign in to access your account." in smaller grey font. There are two input fields: "Username" with the text "admin" and "Password" which is masked with dots. An eye icon is to the right of the password field. At the bottom is a blue "Sign In" button.

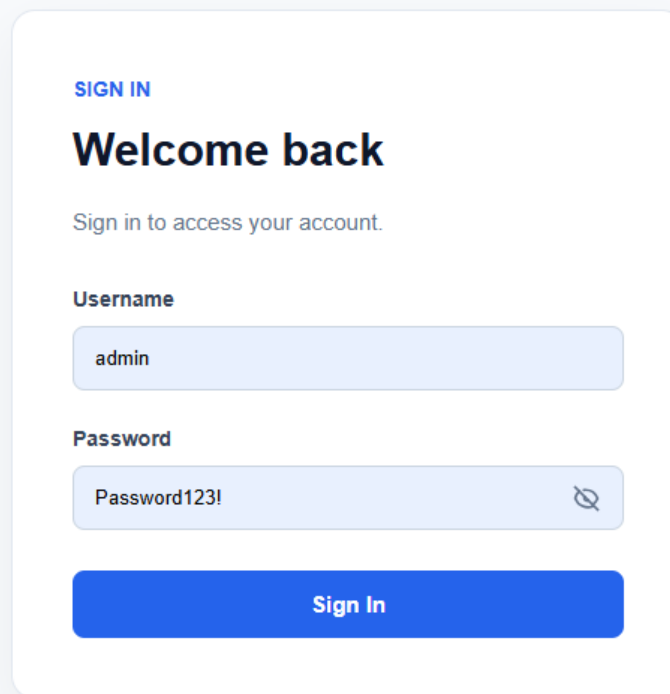
Evidence 02 – Login Page Password Visible State

Related Acceptance Criteria: AC-002, AC-003

Evidence:

Insert a screenshot of the Login page after clicking the eye icon. The password should be visible, and the eye-slash icon should be displayed.

Figure 2: Login page showing the password visibility toggle in the visible state.



The screenshot shows a login form on a light blue background. The form is a white rounded rectangle containing the following elements: a blue 'SIGN IN' link, a 'Welcome back' heading, a subtitle 'Sign in to access your account.', a 'Username' label above a text input field containing 'admin', a 'Password' label above a text input field containing 'Password123!', and a blue 'Sign In' button. An eye icon with a slash is visible on the right side of the password input field, indicating the password is visible.

Evidence 03 – Admin Add Student Form Default Password State

Related Acceptance Criteria: AC-004

Evidence:

Insert a screenshot of the Admin Add Student form showing the password field in its default masked state. The password should appear as dots or bullets, and the eye icon should be visible beside the password input. This screenshot demonstrates that the password is hidden by default when the administrator opens the Add Student form.

Figure 3: Admin Add Student form displaying the password in its default hidden state.

The screenshot displays a modal form titled "Add New Student" with a close button (X) in the top right corner. Below the title is the instruction "Create a new student account." The form contains six input fields arranged in two columns:

- Registration Number:** Contains the text "REG005". Below the field is the hint "Letters, numbers, hyphens, or underscores only."
- First Name:** Contains the text "Kumar". Below the field is the hint "At least 3 characters; letters and spaces only."
- Last Name:** Contains the text "Mala". Below the field is the hint "At least 3 characters; letters and spaces only."
- Email:** Contains the text "Mala@gmail.com". Below the field is the hint "Example: student@example.com".
- Phone Number:** Contains the text "0771234567". Below the field is the hint "Exactly 10 digits."
- Password:** Contains ten dots to mask the password. To the right of the dots is an eye icon. Below the field is the hint "At least 8 characters with uppercase, lowercase, number, and special character."

At the bottom right of the form are two buttons: a light blue "Cancel" button and a blue "Create Student" button.

Evidence 04 – Admin Add Student Form Password Visibility

Related Acceptance Criteria: AC-004, AC-005

Evidence:

Insert a screenshot of the Admin Add Student form showing the password field and the visibility toggle. Demonstrate that the password can be revealed without changing the entered value.

Figure 4: Admin Add Student form displaying the password visibility toggle.

The screenshot displays a modal form titled "Add New Student" with a close button (X) in the top right corner. The form is used to create a new student account and contains the following fields and instructions:

- Registration Number:** Input field containing "REG005". Instruction: "Letters, numbers, hyphens, or underscores only."
- First Name:** Input field containing "Kumar". Instruction: "At least 3 characters; letters and spaces only."
- Last Name:** Input field containing "Mala". Instruction: "At least 3 characters; letters and spaces only."
- Email:** Input field containing "Mala@gmail.com". Instruction: "Example: student@example.com"
- Phone Number:** Input field containing "0771234567". Instruction: "Exactly 10 digits."
- Password:** Input field containing "St1232123@12". To the right of the field is a visibility toggle icon (an eye with a diagonal slash). Instruction: "At least 8 characters with uppercase, lowercase, number, and special character."

At the bottom right of the form are two buttons: a light gray "Cancel" button and a blue "Create Student" button.

6. Final Result

Item	Result
Requirements	Met, subject to verification
Implementation	Completed
Testing	Pass, subject to confirmation of manual tests
Acceptance Criteria	All listed criteria met, subject to verification
Existing Login Functionality	Preserved
Existing Student Creation Functionality	Preserved
Existing CSS Design	Preserved
Final Status	Accepted, subject to testing and evidence review

6.1 Final Summary

CR-016 was implemented as a password visibility enhancement for the Student Management System. The change introduces show/hide password functionality on the Login page and Admin Add Student form.

React state is used to switch the password input between masked and visible text. Eye and eye-slash icons indicate the current visibility state. The toggle button does not submit the form, and the entered password value is preserved when visibility changes.

The existing login submission handler, student creation process, validation rules, error handling, and API integration remain unchanged. Additional CSS rules support the toggle's positioning and appearance without replacing the existing CSS design.

The feature does not require backend or database changes and does not generate additional API requests merely to change password visibility. Final acceptance should be confirmed after executing the test cases and attaching the required screenshots.

6.2 Issues / Limitations

No significant implementation issues are expected.

Password visibility is a frontend display feature. When revealed, the password can be seen by anyone viewing the screen. Users should hide the password after checking it, especially on shared devices or in public environments.

Student Name: _____ A.Paheershana _____

Student ID: _____ JF/ICT/24/04 _____

Date: 9 October 2026

Signature: _____ A.Paheershana _____

~ END OF REPORT ~